



PROJECT-BASED LEARNING

PHASE-I EVALUATION

ASLConnect: Inclusive Communication

Team ID: DSCPP-III-2026-T059

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27



Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T059

Semester: 3rd

Section: A , DS1, ML1

Domain: *Computer Science*

Team Members

1. *Srishti Dhasmana – 2510010598*
2. *Ishita Bijalwan– 2510380152*
3. *Nandini Rana – 2510370024*

Mentor

Mr.Manish Mehta

Interactions completed: 2



Problem Statement & Motivation

Problem Statement

Individuals with hearing impairments often struggle to be part of everyday conversations, while most hearing people lack any knowledge of sign language. This gap exists largely because sign language is seen as difficult to learn, with few structured resources available. Our project aims to bridge this divide by making ASL easier to learn and understand, fostering more inclusive communication for all.

Motivation

- ▶ *No structured path to learn ASL.*
- ▶ *Limits inclusive communication.*
- ▶ *No organised data pattern to learn.*

Target Users / Stakeholders

- ▶ *Family of impaired people.*
- ▶ *Educators, and interpreters in training.*
- ▶ *Faster, structured learning that improves communication*



Objectives

Project Objectives

1. *Develop a web-based solution to simplify ASL learning.*
2. *Implement core features using C++ and DSA concepts.*
3. *Apply DSA logic (searching, sorting, data organization) to structure lessons.*
4. *Test the system for accuracy and performance.*
5. *Demonstrate real-world use and future scope for expansion*

Key Objective: *"Enable users to learn and correctly recognize at least the ASL alphabet and basic words through structured lessons, with measurable improvement in sign accuracy and lesson completion rate."*



Proposed Solution

Proposed Approach

1. *Input / Data Collection*
2. *Processing / Core Algorithm*
3. *System Logic*
4. *Database / Storage*
5. *Output / Visualization*

Key Features

- ▶ *Interactive ASL Lessons*
- ▶ *Text to Sign Matching*
- ▶ *Gesture Recognition and Validation*
- ▶ *Progress Tracking and Scoring*
- ▶ *Visual Feedback*

An interactive website is designed to teach American Sign Language (ASL) using C++ and DSA concepts. User inputs such as text or gestures are collected and processed through efficient algorithms for recognition and matching. The system logic ensures smooth flow and error handling, while structured data storage using arrays, linked lists, and hash maps enables quick retrieval and scalability. Finally, visual outputs like animations and images provide immediate feedback, making ASL learning engaging, accessible, and efficient while reinforcing computational thinking.

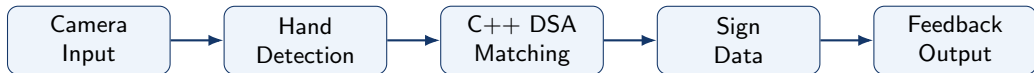


Requirement: Demonstrate meaningful integration of both subjects in **one** project.

Course	Concepts Applied	Project Module
Data Structures in C	<i>Arrays, Linked Lists, Trees, Graphs, Hashing, etc.</i>	Lesson Structuring & Sign Lookup Module
OOPs with C++	<i>Classes, Inheritance, Polymorphism, STL, etc.</i>	Gesture Recognition Engine
Web Design	<i>Styling, Navigation, User Interaction, Content</i>	Website Frontend (HTML/CSS/JS)
Computer Vision	<i>Hand Landmark Detection, Contour Analysis, Real-time Tracking</i>	Gesture Recognition Engine



System Architecture / Workflow



Major Components

- ▶ Website Frontend (HTML/CSS/JS)
- ▶ Computer Vision Module (Hand Detection)
- ▶ C++ DSA Engine (Sign Matching)

Data / Control Flow

- ▶ User selects lesson → website shows sign
- ▶ Webcam captures gesture → landmarks extracted
- ▶ C++ engine matches sign, feedback shown to user



Technology Stack & Methodology

Technology Stack

- ▶ Programming: C++, HTML, CSS, JavaScript
- ▶ Data Storage: JSON / Structured Files
- ▶ Tools / Library: OpenCV (Computer Vision)
- ▶ IDE: VS Code
- ▶ Version Control: Git / GitHub

Development Methodology

1. Problem & Requirement Analysis
2. Designing Lesson Structure & DSA Logic
3. Website & Gesture Engine Implementation
4. Gesture Accuracy Testing
5. Performance & Usability Evaluation

Justify the selection of technologies and methodology.



Initial Progress & Team Contribution

Progress Achieved

- ▶ Background / literature study on ASL learning gaps
- ▶ Requirements identified (lesson structure, gesture recognition)
- ▶ System architecture & workflow designed
- ▶ Website skeleton (HTML/CSS/JS) implemented
- ▶ DSA logic planned for lesson & sign data structuring

Individual Contribution

Member	Role	Contribution
M1	Lead - Frontend and UI	HTML, CSS, JavaScript Interface
M2	DSA and Backend	C++/DSA Implementation, Data Management
M3	CV/ML Developer	OpenCV, ASL Recognition, Dataset



Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ Working ASL learning website with structured lessons
- ▶ Functional gesture recognition for basic ASL signs
- ▶ Faster, more accessible ASL learning for beginners
- ▶ Scope to expand vocabulary & add more sign languages

References

1. ASL Alphabet Dataset – M (grassknotted/asl-alphabet)
2. MediaPipe Hands – Google, hand landmark detection framework
3. OpenCV documentation – opencv.org
4. A. Amin et al., "Open-Source ASL Fingerspell Recognition and Semantic Pose Retrieval Interface," arXiv:2408.09311

Thank You
Questions & Suggestions